

Total No. of Questions : 8]

SEAT No. :

PC-2507

[Total No. of Pages : 2

[6354]-637

B.E. (Mechanical Engineering)

ELECTRICAL AND HYBRID VEHICLE

(2019 Pattern) (Semester - VIII) (402051 E) (Elective - VI)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of electronic pocket calculator is allowed.
- 5) Assume Suitable data if necessary.

Q1) a) Explain and estimate the Battery Performance Parameters. [8]

b) Explain power efficiency and characteristics curve of Lithium-ion battery. [10]

OR

Q2) a) A rickshaw has following specifications : [12]

Grass curb weight (GCW) = 600 kg

Road friction, $R_{\{r\}} = 0.015$

Coefficient of drag $C_{\{d\}} = 0.45$

Density of air $\rho = 1.2 \text{ kg / m}^3$

Frontal area of Vehicle, $A = 1.6 \text{ m}^2$

Wheel radius = 0.2m

Maximum speed of the vehicle = 30 kmph (8.34 m/s)

Working voltage of motor = 24 Volts

For attaining the speed range of 90 km per charge, suggest the type and capacity of battery pack required. Assume suitable data wherever required.

b) Explain Units of Battery/Fuel Cell Energy Storage. [6]

P.T.O.

Q3) a) Describe and illustrate various electric drive-train topologies and their types for Four-Wheeler application. **[9]**

b) Describe the Basic concept of Electric Traction with examples. **[8]**

OR

Q4) a) Describe and Illustrate the Brake System and its types. **[9]**

b) Explain Power flow Control in Electric Drive-Train Topologies. **[8]**

Q5) a) Describe and Illustrate the types of Electric Vehicle frame Configurations. **[9]**

b) Describe Indian AIS Standards for e-Vehicles. **[9]**

OR

Q6) a) What is the homologation of vehicles? **[9]**

b) Describe and illustrate the Driving dynamics and Comfort. **[9]**

Q7) a) Describe and illustrate Charger Architectures. **[9]**

b) Describe AIS Charging Standards. **[8]**

OR

Q8) a) Explain the Level 1, Level 2 and Level 3 chargers of electric vehicles. **[9]**

b) Explain BIS Charging Standards. **[8]**

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